

Kartik Kailas Thakare

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Education

- Rochester Institute of Technology**, MS in Electrical Engineering Aug 2024 – May 2027
- **Coursework:** Advanced Design of Digital Systems, Design of Computer Systems, Complex Digital Systems Verification.
- University of Mumbai**, BE in Electronics Engineering Aug 2019 – May 2023
- **Coursework:** Digital Logic Circuits, Basic VLSI Design, Embedded Systems & RTOS, DSP, Machine Learning.

Experience

- Embedded Firmware Engineer Co-Op**, Skyworks Solutions, Inc. – Hillsboro, Oregon June 2026 – Present
- Developed and validated embedded firmware for wireless communication products using C/C++, Git, Python and Linux.
 - Performed firmware testing and debugging for releases using industry-standard debugging and validation tools.
- Graduate Research Assistant**, Rochester Institute of Technology – Rochester, New York Aug 2025 – May 2026
- Applied electromagnetic theory and transmission line concepts to analyze signal behavior and system-level effects.
 - Developed and refined technical documentation, supporting instructional and reference material.
- EME General Laboratory Assistant**, Rochester Institute of Technology – Rochester, New York May 2025 – May 2026
- Maintained hardware bring-up and validation by setting up, calibrating, and troubleshooting lab equipment (oscilloscopes, logic analyzers, power supplies, multimeters) assisting in root-cause analysis of system issues.
 - Managed lab operations by organizing equipment, managing inventory, and preparing course-specific kits.
- Systems Engineer**, Tata Consultancy Services – Mumbai, India Oct 2023 – June 2024
- Worked on Linux/Oracle enterprise systems, contributing to stability, upgrades, migrations, and production deployments.

Projects

- ASIC Design Verification (UVM & SystemVerilog) – RCC Block of DTMF Receiver** Dec 2025
- Achieved 96% full digit-level coverage closure using SVA, functional/FSM/cross coverage, and watermark-driven simulation control; analyzed results with Cadence Xcelium & IMC.
 - Developed a UVM-based verification environment (agent, driver, monitor, scoreboard) for the RCC block, validating functionality across RTL and gate-level netlist simulations.
 - Integrated a C/C++ reference model via SystemVerilog DPI and TLM FIFOs for accurate scoreboard-based DUT-vs-model checking with detailed mismatch diagnostics.
- Design & Verification of Multi-Channel ADPCM Codec** May 2025
- Delivered an 8-channel ITU-T G.726/G.711 compliant voice codec (16–40 kbit/s) by implementing a bit-serial APRSC RTL architecture for area-efficient DSP processing.
 - Developed Verilog datapath and FSM control for bit-serial ACCUM/FMULT pipeline within a Wishbone-based SoC.
 - Verified functionality using directed/random simulations with ITU vectors and completed synthesis, DFT insertion, and static timing analysis using Cadence/Synopsys tools.
- 15-bit Pipelined RISC Processor with 2-Way Set-Associative Cache and an Assembler** Nov 2025
- Designed a 5-stage pipelined 15-bit RISC Processor with hazard handling, forwarding, Von Neumann-based, memory-mapped I/O, and 2-way set-associative cache; implemented RTL in Verilog.
 - Built a Python assembler and scaled the processor to 2-core and 4-core systems to perform 4x4 matrix multiplication, achieving 3–4× speedup measured using cycle counters; verified using ModelSim and prototyped on Cyclone-V FPGA.
- Parameterized RTL Code & Testbench Generator Using Perl & Python** Sept 2025
- Developed a configurable RTL generator producing synthesizable Verilog and testbenches.
 - Automated multi-stage pipeline register generation and validated outputs in Cadence EDA flows.
- Memory Access Bus Arbiter & DTMF Receiver Design with Verilog HDL.** Nov 2024
- Designed a shared memory bus arbiter for TDSP–DMA access and integrated it with a DTMF receiver core.
 - Performed RTL and gate-level simulation, STA, and power analysis to validate timing and functionality.

Publications

- A System and Process for Identifying and Tracking Movements of a Badminton Player** Oct 2023
- Pawan Wankhede, *Kartik Thakare*, Bhumika Thombare, Navin Singhaniya
- [Indian Patent Journal 202321058914 A](#)

Skills

Digital Design & Verification: RTL Design, ASIC Verification, UVM, Assertions (SVA), Functional & Code Coverage, Pre-Silicon Validation, Physical Design, Place & Route, DRC, LVS, Clock Tree Synthesis (CTS), Static Timing Analysis (STA)

FPGA & Computer Architecture: FPGA Prototyping, Pipelined CPUs, Cache & Memory Hierarchy, Memory-Mapped I/O, Multicore Systems, FSM, UART, SPI, I2C, RTOS, DMA, Cache Coherency, Multi-clock Domain Design, FPGA IP Integration

Languages: Verilog, SystemVerilog, Perl, Python, C, C++, Assembly

Technologies: Cadence Virtuoso, Quartus Prime, Xilinx Vivado, SPICE, ModelSim, Altium, Git, Synopsys, Linux